

CS2206: Computer architecture

Lab 10: Study of MIPS Pipelined Processor implementation

_Task 1: Study the given pipeline implementation of the processor, and identify error in the design (if any) . Convert the following C code to MIPS assembly, test the code in MIPS pipelined version(Given) .

```
for(i = 0; i < 100; i++)
```

```
A[i] = i+1;
```

Task 2: *Write ASCII values of your name full name to the memory using appropriate instruction, and write program to find the sum of those values using the given pipeline MIPS design. Compute the number of cycles required and compare with number of cycles that you got in previous(Multi-cycle implementation).* Enter the relevant information in the record for demonstration.

Task 3 : Design and implement two CSE algorithms in MIPS assembly, run them on the specified pipelined processor, and compute their CPI.